

D470

Transportation, Logistics, and Distribution

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A medical device maker must get 40 replacement heart-monitor parts from Memphis to a hospital in Denver by tomorrow morning; a patient procedure depends on them. The shipping manager holds a CLTD certification. Which mode fits?

- A. Expedited air freight.
- B. Standard truckload.
- C. Rail, because it is the lowest-cost option.
- D. Intermodal rail and truck.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A furniture importer in Chicago buys from a supplier in Vietnam under FOB Ho Chi Minh City. The container was loaded by a crew working a double shift. The vessel is damaged in a storm mid-ocean. Who bears the risk of loss?

- A. The seller, until the goods reach Chicago.
- B. The buyer, because risk passed when the goods were loaded on board in Vietnam.
- C. Buyer and seller split the loss equally.
- D. The ocean carrier, always.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Under which Incoterm does the seller bear the most cost and risk, delivering to the buyer's named place with import duties paid?

- A. FOB
- B. EXW
- C. CIF
- D. DDP

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A retailer compares two routes for 1,000 units worth \$200 each. Route A: freight \$3,000 and 30 days in transit. Route B: freight \$6,000 and 5 days in transit. Inventory carrying cost is 36% a year (use a 360-day year). The Route B carrier just repainted its fleet. Which route has the lower total cost?

- A. Route A, because a longer transit spreads the cost.
- B. Route B, saving \$5,000.
- C. Route B: \$7,000 vs. \$9,000.
- D. Route A, because its freight is half the price.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A shipment costs \$200 per day in transit to carry. Route A has \$3,000 freight and 30 days in transit. Route B takes 5 days. What is the most Route B can charge in freight and still tie Route A's total cost?

- A. \$5,000
- B. \$8,000
- C. \$3,000
- D. \$9,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A grocery distributor with a 50-year history receives full truckloads of produce from 12 growers each morning and must send mixed loads to 80 stores the same day. Its warehouse is 400,000 square feet. Which strategy minimizes handling and storage?

- A. Put produce away in reserve racks, then pick orders later.
- B. Direct delivery from each grower to each store.
- C. Add storage, since the building is large.
- D. Cross-docking.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A hospital uses 40 boxes of gloves a day. Supplier lead time is 5 days, and the materials manager keeps 60 boxes of safety stock. The hospital has 320 beds. At what inventory level should it reorder?

- A. 320 boxes
- B. 200 boxes
- C. 260 boxes
- D. 140 boxes

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

8. [TRANSFER — reversed stem / different variable]

A warehouse reorders at 900 units. Daily demand is 120 units and lead time is 6 days. How much safety stock does this imply?

- A. 180 units
- B. 1,620 units
- C. 0, since reorder points exclude safety stock
- D. 720 units

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

9.

A parts distribution center, recently expanded by 30,000 square feet, finds that 15% of its SKUs account for 70% of picks. Pickers walk an average of 9 miles per shift. Where should those SKUs go?

- A. On top rack levels to save floor space.
- B. In the new expansion space.
- C. In the most accessible, waist-high locations nearest shipping.
- D. Alphabetically by part number.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

10.

A shipper pays \$0.25 per pound for LTL or a flat \$2,500 for a full truckload. The shipper's dock has four doors. Above what shipment weight is the full truckload cheaper?

- A.** 10,000 pounds
- B.** 2,500 pounds
- C.** 625 pounds
- D.** Never; LTL always costs less for partial loads.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A 12,000-pound shipment is quoted \$2,700 as a full truckload. What LTL rate per pound would make the two options cost the same?

- A.** \$0.27 per pound
- B.** \$0.225 per pound
- C.** About \$4.44 per pound
- D.** \$0.225 per hundredweight

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A carrier picks up 24 pallets of electronics from a manufacturer known for fast innovation. At delivery, the consignee counts 22 pallets. Which document is the key evidence of what the carrier received and the terms of carriage?

- A.** The purchase order.
- B.** The certificate of origin.
- C.** The commercial invoice.
- D.** The bill of lading.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A medical device maker must get 40 replacement heart-monitor parts from Memphis to a hospital in Denver by tomorrow morning; a patient procedure depends on them. ~~The shipping manager holds a CLTD certification.~~ Which mode fits?

Target word: mode **Decides:** Overnight deadline for small, critical goods.

A	Correct	Only air reliably meets an overnight, long-distance deadline.
B	Adjacent category	About 1,000 miles is too far for a standard truck overnight, and 40 parts are far too few for a truckload.
C	Inverted operation	Optimizes cost when time decides.
D	Confusion set	Intermodal is cheaper but slower still.

2. Answer B Concept 2

A furniture importer in Chicago buys from a supplier in Vietnam under FOB Ho Chi Minh City. ~~The container was loaded by a crew working a double shift.~~ The vessel is damaged in a storm mid-ocean. Who bears the risk of loss?

Target word: risk bearer **Decides:** Under FOB, risk passes when goods are loaded on board at the origin port.

A	Inverted operation	That describes a destination term such as DAP or DDP.
B	Correct	FOB transfers risk at the ship's rail, as goods go on board at origin.
C	Adjacent category	Incoterms assign risk to one party at a defined point.
D	Absolute language	Carrier liability is limited; Incoterms decide buyer vs. seller risk.

3. Answer D Concept 2 · transfer

Under which Incoterm does the seller bear the most cost and risk, delivering to the buyer's named place with import duties paid?

Target word: reverse: description to term **Decides:** Delivered, duties paid.

A	Adjacent category	Risk passes at loading on board at the origin port.
B	Inverted operation	Ex Works puts maximum obligation on the buyer.
C	Confusion set	The seller pays freight and insurance, but risk passes at loading.
D	Correct	Delivered Duty Paid puts maximum obligation on the seller.

4. Answer C Concept 3

A retailer compares two routes for 1,000 units worth \$200 each. Route A: freight \$3,000 and 30 days in transit. Route B: freight \$6,000 and 5 days in transit. Inventory carrying cost is 36% a year (use a 360-day year). ~~The Route B carrier just repainted its fleet.~~ Which route has the lower total cost?

Target word: lower total cost **Decides:** Carrying cost = $\$200,000 \times 0.36 / 360 = \200 per day in transit.

A	Inverted operation	Longer transit adds carrying cost.
B	Wrong stem number	Counts only the carrying-cost gap and ignores B's \$3,000 higher freight.
C	Correct	A: $3,000 + 30 \times 200 = 9,000$; B: $6,000 + 5 \times 200 = 7,000$.
D	Dropped element	Drops in-transit carrying cost.

5. Answer B Concept 3 • transfer

A shipment costs \$200 per day in transit to carry. Route A has \$3,000 freight and 30 days in transit. Route B takes 5 days. What is the most Route B can charge in freight and still tie Route A's total cost?

Target word: solve for break-even freight **Decides:** A total = \$9,000; B carrying = \$1,000.

A	Dropped element	Only the carrying-cost gap; drops A's freight.
B	Correct	$9,000 - 5 \times 200 = 8,000$.
C	Wrong stem number	Route A's freight only.
D	Adjacent category	Route A's total; ignores B's own carrying cost.

6. Answer D Concept 4

A grocery distributor ~~with a 50-year history~~ receives full truckloads of produce from 12 growers each morning and must send mixed loads to 80 stores the same day. ~~Its warehouse is 400,000 square feet.~~ Which strategy minimizes handling and storage?

Target word: flow strategy **Decides:** Same-day inbound-to-outbound mixing.

A	Adjacent category	Adds storage and handling for goods that ship the same day.
B	Confusion set	Up to 960 small shipments instead of consolidated loads.
C	Camouflage	Size does not change the same-day flow.
D	Correct	Sorts inbound loads straight to outbound trucks with little storage.

7. Answer C Concept 5

A hospital uses 40 boxes of gloves a day. Supplier lead time is 5 days, and the materials manager keeps 60 boxes of safety stock. ~~The hospital has 320 beds.~~ At what inventory level should it reorder?

Target word: reorder point **Decides:** $40 \times 5 + 60$.

A	Camouflage	The bed count, not an inventory level.
B	Dropped element	Leaves out safety stock.
C	Correct	200 lead-time demand + 60 safety stock.
D	Inverted operation	Subtracts safety stock.

8. Answer A Concept 5 • transfer

A warehouse reorders at 900 units. Daily demand is 120 units and lead time is 6 days. How much safety stock does this imply?

Target word: solve for safety stock **Decides:** $900 - 120 \times 6$.

A	Correct	$900 - 720 = 180$.
B	Inverted operation	Adds lead-time demand instead of subtracting.
C	Absolute language	Safety stock is part of the reorder point.
D	Confusion set	Lead-time demand, not safety stock.

9. Answer C Concept 6

A parts distribution center, ~~recently expanded by 30,000 square feet~~, finds that 15% of its SKUs account for 70% of picks. Pickers walk an average of 9 miles per shift. Where should those SKUs go?

Target word: slotting **Decides:** A few SKUs drive most picks.

A	Inverted operation	Makes the most-picked items the hardest to reach.
B	Camouflage	New space is not necessarily close to shipping.
C	Correct	Cuts travel on the most frequent picks.
D	Adjacent category	Easy to find, but ignores pick frequency.

10. Answer A Concept 7

A shipper pays \$0.25 per pound for LTL or a flat \$2,500 for a full truckload. ~~The shipper's dock has four doors.~~ Above what shipment weight is the full truckload cheaper?

Target word: break-even weight **Decides:** \$2,500 / \$0.25 per pound.

A	Correct	$2,500 / 0.25 = 10,000$.
B	Wrong stem number	The truckload price read as a weight.
C	Inverted operation	Multiplies \$2,500 by the rate.
D	Absolute language	Heavy partial loads can cost more by LTL.

11. Answer B Concept 7 • transfer

A 12,000-pound shipment is quoted \$2,700 as a full truckload. What LTL rate per pound would make the two options cost the same?

Target word: solve for rate **Decides:** \$2,700 / 12,000 pounds.

A	Wrong stem number	Misplaces the decimal from \$2,700.
B	Correct	$2,700 / 12,000 = 0.225$.
C	Inverted operation	$12,000 / 2,700$ divides the wrong way.
D	Confusion set	Right number, wrong unit; a hundredweight is 100 pounds.

12. Answer D Concept 8

A carrier picks up 24 pallets of electronics from a manufacturer ~~known for fast innovation~~. At delivery, the consignee counts 22 pallets. Which document is the key evidence of what the carrier received and the terms of carriage?

Target word: document **Decides:** The carrier's signed receipt and contract.

A	Adjacent category	The buyer's order; it says nothing about what the carrier took.
B	Confusion set	Shows where goods were made, not what was shipped.
C	Confusion set	A seller-to-buyer billing document; the carrier does not sign it.
D	Correct	Signed by the carrier at pickup; it records the pallet count and contract terms.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Mode selection	Match mode to the need: air for urgent, high-value, light goods; truck for flexible door-to-door; rail and ocean for low cost on heavy, non-urgent freight.	Choosing the cheapest mode when time decides.	A hard deadline or a cost-driven bulk move.
2	Incoterms and risk transfer	Each Incoterm fixes where risk passes: FOB at loading on board at origin; DDP at the buyer's door with duties paid; EXW at the seller's premises.	Assuming the seller bears risk until the goods arrive.	An Incoterm plus loss or damage in transit.
3	Total landed cost	Compare freight plus in-transit carrying cost (and other adders), not freight rates alone.	Picking the lowest freight quote.	A cheap slow option vs. a costly fast one, with inventory value.
4	Cross-docking	Cross-docking moves inbound goods straight to outbound loads with little or no storage.	Putting fast-flowing goods away in storage first.	Full inbound loads that must go out mixed the same day.
5	Reorder point	Reorder point = daily demand × lead time + safety stock.	Leaving out safety stock or subtracting it.	Daily usage, lead time, and a safety stock figure.
6	ABC slotting	Put the fastest-moving items in the most accessible spots nearest shipping.	Slotting alphabetically or in whatever space is new.	A small share of SKUs driving most picks.
7	LTL vs. FTL break-even	FTL becomes cheaper when weight × LTL rate exceeds the flat truckload charge.	Multiplying the truckload charge by the rate.	A per-pound LTL rate and a flat FTL price.
8	Bill of lading	The bill of lading is the carrier's receipt for the goods and the contract of carriage (and can serve as a document of title).	Relying on the invoice or purchase order as proof of what the carrier took.	A count dispute between pickup and delivery.

Course scope (public description): Transportation, Logistics, and Distribution examines logistics and the planning and management of transportation and distribution. This course introduces requirements and risks, facilities and inventory, strategy and supply chain synchronicity, efficiencies and costs, and laws and regulations related to transportation, distribution, and logistics. Students will explore warehousing and warehouse management, inventory and logistics management, distribution, and supply chain management from the lens of transportation, logistics, and distribution. This course is aligned with the Certified in Logistics, Transportation, and Distribution (CLTD) certification from the Association for Supply Chain Management.